

## Multiple Choice (10 marks)

1. What pressure is required to compress 5 liters of gas at 1 atm. pressure to 1 liter at a constant temperature?
  - (a) 3 atm
  - (b) 9 atm
  - (c) 5 atm
  - (d) 6 atm
  - (e) 10 atm
2. Calculate the percentage difference between  $e^x$  and  $1 + x$  for  $x = 0.0050$ 
  - (a)  $7.50 \times 10^{-3}\%$
  - (b)  $2.50 \times 10^{-3}\%$
  - (c)  $0.75 \times 10^{-3}\%$
  - (d)  $2.00 \times 10^{-3}\%$
  - (e)  $1.25 \times 10^{-3}\%$
3. A stock solution of 12.0 M sulfuric acid is made available. What is the best procedure to make up 100. mL of 4.0 M sulfuric acid using the stock solution and water prior to mixing?
  - (a) Add 67.7 mL of 12.0 M acid to the flask, and then dilute it with 33.3 mL of water.
  - (b) Add 33.3 mL of water to the flask, and then add 66.7 mL of 12.0 M acid.
  - (c) Add 40 mL of water to the flask, and then add 60 mL of 12.0 M acid.
  - (d) Add 60 mL of 12.0 M acid to the flask, and then dilute it with 40 mL of water.
  - (e) Add 50 mL of 12.0 M acid to the flask, and then dilute it with 50 mL of water.
4. Assuming complete ionization, calculate (a) the pH of 0.0001 NHCl, .(b) the pOH of 0.0001 N KOH.
  - (a) pH is 10, pOH is 4
  - (b) pH is 4, pOH is 10
  - (c) pH is 3, pOH is 3
  - (d) pH is 4, pOH is 4
  - (e) pH is 1, pOH is 1
5. Calculate the energy of a hydrogen atom in its ground state ( $n = 1$ ) using the mass of the electron rather than the reduced mass of the hydrogen atom.

- (a) - 13.61 eV
- (b)  $1.6022 \times 10^{-19}C$
- (b) -  $2.1802 \times 10^{-18}J$
- (b)  $9.1095 \times 10^{-31}kg$

## True / False (10 marks)

*Answer True or False*

6. True or False: The reaction of 2 g of VO with 5.75 g of  $Fe_2O_3$  can yield 2.18 g of  $V_2O_5$  as a product.
7. True or False: The atomic weight of Calcium is approximately 40.2 g/mol, based on the amount of charge liberated during electrolysis.
8. True or False: An ester is formed through the reaction of an alcohol and a carboxylic acid.
9. True or False: The solutions to the equation  $x^2 - 7x + 10 = 0$  are  $x = 3$  and  $x = 4$ .
10. True or False: London dispersion forces arise from temporary dipoles formed by the fluctuating positions of electrons in a molecule.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Given a compound composed of 51.9% carbon, 4.86% hydrogen, and 43.2% bromine, calculate its empirical formula and discuss the significance of empirical formulas in chemical analysis.
12. Calculate the total volume of water lost by an average adult over 60 years through urine, sweat, fecal matter, and expired air, discussing the implications for hydration and physiology.

*End of Paper*